SAMPLE QUESTIONS 2A1 (MULTIPLE CHOICE)

(NOTE: These questions are intended as representations of the style of questions that may appear on examinations, they are not intended as study material and, as such, may not be in line with any current examination syllabus)

Currently all the 2A1 SOPEEC examinations are delivered in a multiple-choice format. Examination candidates have 3.0 hours to answer 100 multiple choice questions. This works out to be 1.8 minutes per question.

To use your time wisely it is recommended that you read through the questions and answer only the questions for which the answer comes to you quickly. Leave other questions and calculation questions for later.

What is the maximum power input of an electric boiler for it to require only one pressure relief valve?

a. 565 kW
b. 747 kW
c. 1,100 kW
d. 1,500 kW

In which of the corrugation furnace designs currently recognized in ASME Section I (when C = a constant, the value of which depends on the type of furnace) does C = 108 when corrugations are not more than 200 mm from centre to centre and not less than 32 mm deep?

a. Fox
b. Plain
c. Brown
d. Morrison

Which type of law may be defined as private law, which exists between parties?

a. Civil law
b. Regulatory law
c. Constitutional law
d. Administrative law

What type of site-specific safety training is normally provided for contractors, temporary workers, and visitors?

a. Safe work plan
b. Operational review
c. Initial status review
d. Risk reduction plan

Hazard identification is a systematic process used to identify any condition within the workplace that has the potential to cause

a. injury but does not include property damage or environmental impact.
b. property damage and its impact on the worker during non-emergency situations.
c. injury, property damage, or environmental impact including emergency situations
d. injury and property damage but does not include environmental impact and non-emergency situations

A flywheel rotating at 250 RPM is sped up at 600 RPM over a period of 8 seconds. Determine the average angular velocity during the 8 seconds.

a. 26.18 rads/second
b. 44.51 rads/second
c. 52.36 rads/second
d. 62.80 rads/second

A force of 800 N is applied at a radius of 750 mm to turn a shaft. If the power transmitted is 25 kW find the RPM of the shaft.

a. 97 RPM
b. 132 RPM
c. 399 RPM
d. 527 RPM

What type of load increases and decreases in magnitude many times over specific periods of time?

a. Cyclic load
b. Static load
c. Impact load
d. Sustained load

The indicated power for an engine is calculated at 8,500 kW. Brake power is measured as 8,000 kW. What is the mechanical efficiency of the engine?

a. 80%
b. 85%
c. 94%
d. 98%

What is the angle formed at the center of the pulley between the perpendiculars drawn to the center from the belt tangents on opposite sides of the pulley?

a. Radian
b. Wrap angle
c. Torque angle
d. Friction angle

When considering rotating masses, the radius at which the total mass is assumed to be concentrated is known as

a. Wheel radius
b. Moment of inertia
c. Angular velocity
d. Radius of gyration

A tank contains a 4,700 kg mass of oil that has a relative density of 0.86. What is the volume of the oil?

a. 3.92 m^3
b. 4.03 m^3
c. 4.12 m^3
d. 5.47 m^3